

RoadSense AI — PRD Review (Summary)

A one-page condensation of [PRD_Review_and_Actions.md](#) . Full detail, per-section reasoning, and the Master Action List live in that document.

Date: 2026-07-18 · Sections reviewed: **all 20**

The headline

The software is built. Almost nothing is proven in the real world.

Platform $\approx$ 90% done · AI $\approx$ 40% done · Overall $\approx$ 75%. Every feature exists in code; the gaps are **real-world validation** (a moving vehicle, real data volume, a stopwatch) and the **AI model** (2 of 7 classes, accuracy targets falsified).

Section-by-section verdict

#	Section	Verdict
1	Goals	Vague/unmeasurable → made them numeric; split "we control" vs "impact".
2	Target Users	5 users → really 3 (Authorities, Fleets, Admin). Citizens dropped.
3	System Overview	Flow OK, but hides the big choice → cloud now, edge later.
4	Hardware	Pi/Jetson is future edge gear. Beta device = smartphone.
5	AI Pipeline	5 of 7 steps done. Remove segmentation (proven harmful); depth = post-beta.
6	Detection Classes	1.5 of 7 built. Cut to 4 + water flag; drop speed breaker + patch repair.
7	Severity	Only 1 of 5 factors used. Severity = size (+depth later); split out Priority ; drop speed.
8	Verification	Strong, but "verified" = seen once. Require ≥2 sightings.
9	Data Captured	9 of 10 fields captured. Decide weather ; add GPS source.
10	Backend Stack	Accurate. Rename → Tech Stack ; add map, Docker.
11	Dashboards	All 11 built — strongest section. Missing: notification bell, weather.
12	Repair Workflow	Code = PRD exactly. Rename "Verified" (clashes); no real authority has used it.
13	APIs	All 7 exist, but list is stale — missing whole features. Regroup by feature.
14	Security	3 solid, 2 shaky. GPS check was silently broken. Missing: privacy (blur faces/plates).
15	Functional Reqs	Mostly met. Not truly "real-time" (~10 fps). Finish the bell.
16	Non-functional Reqs	The honesty crunch. FP rate smashed (0%); precision falsified; 3 untested.
17	Future Features	Road Health Score already built (misfiled). Traffic-sign/accident = different product.
18	Roadmap	4 empty labels. Add "done-when" lines; sync from CLAUDE.md.
19	Success Metrics	Duplicate of Non-functional. Merge them.
20	Long-Term Vision	Good. "Crowdsourced" → "fleet vehicle data".

What's genuinely strong

- **Zero false positives** (0/57) on real footage — the best result in the project.
- **Duplicate clustering** (verification engine) — the real moat.
- **All 11 dashboard features** built and working.
- **Repair workflow** — code matches the PRD exactly; a thoughtful state machine.

What's genuinely weak

- **Accuracy targets are falsified** — 95% precision unreachable by scaling (ceiling = label quality).
- **Only 2 of 7 classes** — and crack barely fires on real video.
- **No validation from a moving vehicle** — every result is out-of-domain (handheld footage).

- **Privacy is absent** — street photos capture faces + plates; a legal blocker for deployment.
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The 4 decisions that unblock everything

1. **Re-baseline accuracy targets** (D-4) — replace 95% precision with an honest, measured number.
 2. **Pick the #1 user** — Authority (buys/fixes) vs Fleet (drives/generates data).
 3. **Confirm cloud-now / edge-later** — and write it into the PRD.
 4. **Lock class scope 7 → 4** — pothole, crack, broken road, edge damage (+ water flag).
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The 1 action that unblocks the most

Get real dashcam footage from a moving vehicle. It clears the accuracy re-baseline *and* the biggest untested risk in one move.

Full reasoning, code references, and the complete 4-bucket action list (Decisions · PRD edits · Code · Tests) are in [PRD_Review_and_Actions.md](#).
